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- ➡ Orthodontics includes the study of growth and development of the jaws and face , and the body generally as influencing the position of the teeth



# Services offered by the Orthodontist

- Preventive Orthodontics
  - Interceptive Orthodontics
  - Corrective Orthodontics
  - Surgical Orthodontics
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## ➤ Preventive Orthodontics

- “actions taken to preserve the integrity of what appears normal for that age”.

## ➤ Interceptive Orthodontics

- “Science and art of Orthodontics employed to recognize and eliminate potential irregularities in the developing dentofacial complex”

## ➤ Corrective Orthodontics

- “Recognizes the existence of a malocclusion and the need for employing certain technical procedures to reduce or eliminate the problem”

## ➤ Surgical Orthodontics

- “It is the combined Orthodontic and Surgical management of severe dento-facial deformities that cannot be treated by Orthodontic therapy alone”



# Aims of Orthodontic treatment

- Functional efficiency
  - Structural balance
  - Esthetic harmony
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## ➤ **Functional Efficiency**

- The orthodontic treatment should aim at improving the functioning of the oro-facial structure

## ➤ **Structural Balance**

- dentoalveolar system
- the skeletal tissue
- the soft tissue including musculature
- Stable orthodontic treatment is best achieved by maintaining a balance between these three tissue systems

## ➤ **Esthetic harmony**

- Many malocclusions can affect the self image of the individual.
- Orthodontic treatment should aim at improving the esthetics of the individual

# Unfavourable sequelae of malocclusion

## ➤ Poor facial appearance

- Malocclusion is adversely affecting the facial appearance of an individual.

## ➤ Risk of caries

- Mal aligned teeth makes the oral hygiene maintenance a difficult task, thereby increasing the risk of caries.

## ➤ Predisposition to periodontal disease

- Malocclusion with poor oral hygiene leads to periodontal diseases.
- Teeth in traumatic occlusion results in damage to periodontal tissues.

## ➤ Risk of trauma

- Teeth that are severely proclined are often at a high risk of injury either during games or by accidental fall.



- **Psychological disturbances**

- Malocclusion affecting adversely the appearance ,makes a person highly self conscious and turns him into an introvert.

- **Abnormalities of function**

- improper swallowing, defects in speech ,improper respiration,

- **Temporomandibular joint problems**

- Malocclusion associated with occlusal prematurities and deep bite can cause TMJ problems such as pain and dysfunction.

- **Interference to the normal growth and development**

- The growth of maxilla is restricted by the mandible in skeletal class 3 malocclusion with anterior crossbite.

# *Need for orthodontic treatment*

- To improve the **aesthetics** of an individual.
- To reduce the susceptibility to dental **caries and periodontal diseases**
- To reduce the susceptibility to **accidental injury**
- To correct the **abnormal muscle activity**
- To correct the **abnormal oral habits**
- To intercept and correct the **interferences to the normal growth**
- To manage the **temporomandibular joint problems**
- To facilitate the **prosthetic rehabilitation**
- To correct **the speech defects**



## **Scope of orthodontic treatment**

Orthodontic treatment can bring changes in the dentition ,the skeletal system and the enveloping soft tissue.

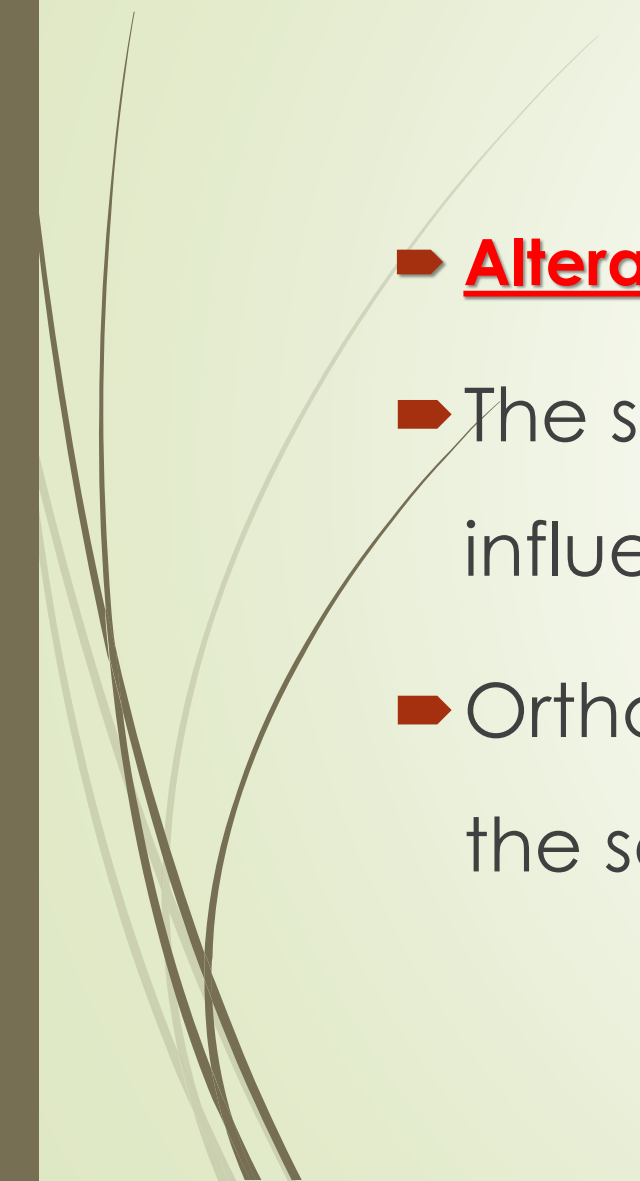
- ➡ Alteration in tooth position
- ➡ Alteration in skeletal pattern
- ➡ Alteration in soft tissue pattern

## ■ Alteration in tooth position

Orthodontic treatment is made possible by the fact that teeth can be moved through the bone to ideal locations by applying appropriate force on them.

## ■ Alteration in skeletal pattern

- Malocclusion may associated with skeletal disharmony of jaw bones.
- Deviations from normal can arise in **Size Position Relationship** between these skeletal components.
- Orthodontist can use orthopaedic forces to bring changes in sagittal, vertical and transverse planes.



## ➤ Alteration in soft tissue pattern

- The soft tissues that envelope the dentition are greatly influenced by the position of dentition.
- Orthodontic treatment can bring favourable changes in the soft tissue pattern.

# ***TERMINOLOGY***





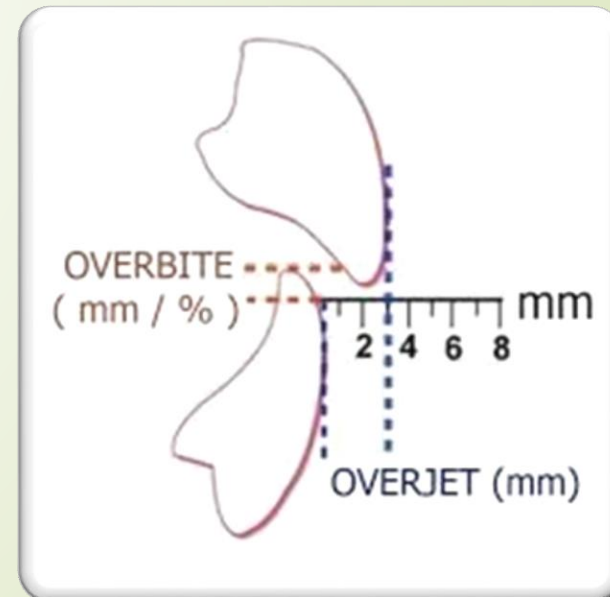
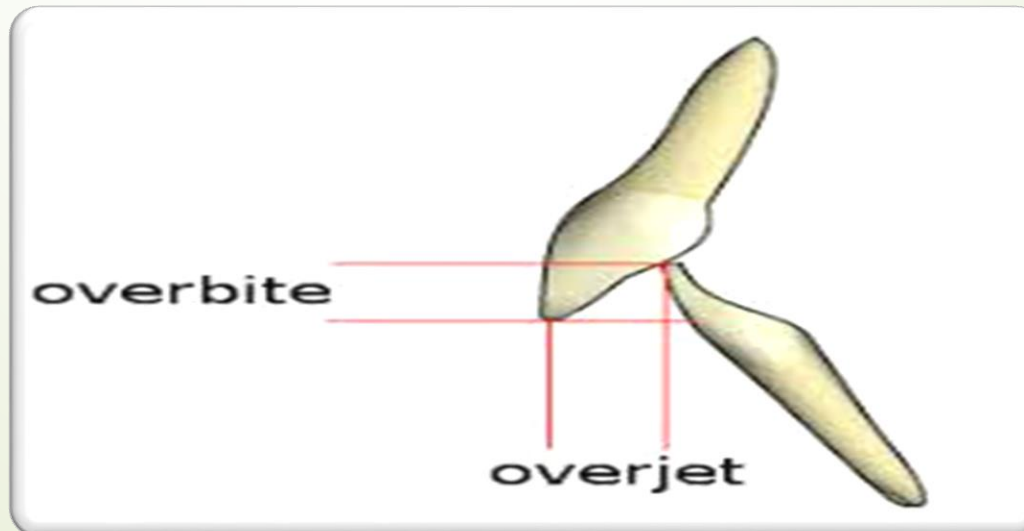
# Orthodontics:

It refers to the straightening & arrangement of the teeth in a proper alignment

## Over bite

the overlap of lower incisors by the upper incisors in the vertical plane

**Normal overbite is 2-3mm or 30% percent or 1/3 rd the clinical crown height of the mandibular incisors.**



## Deep bite:

The increased vertical overlap of lower incisors by the upper incisors more than  $\frac{1}{3}$  or 30 % of the crown of lower incisor is called 'deep bite'.



## Complete over bite:

- which the lower incisors contact either the upper incisors or the palatal mucosa. There is often trauma to the palatal gingiva with pockets, gingival enlargement, recession & food packing.



## Incomplete over bite:

- ➔ in which the lower incisors do not contact either the upper incisor or the palatal mucosa.



## **Closed – deep Bite:**

- The lower incisors are not visible (closed) .
- in Angle's Class II div 2 malocclusion
- or from loss of posterior teeth as a result of over closure of the mandible.



## Edge to Edge bite:

- The upper and lower incisors meet upon their incisal edges when the teeth are in habitual/centric occlusion.

**Both the over bite as well as the over jet will be zero**



## Anterior open bite:

- The Lower incisors are not overlapped in the vertical plane by the upper incisors
- A gap/space is present between the incisal edges of the upper & lower anterior teeth.



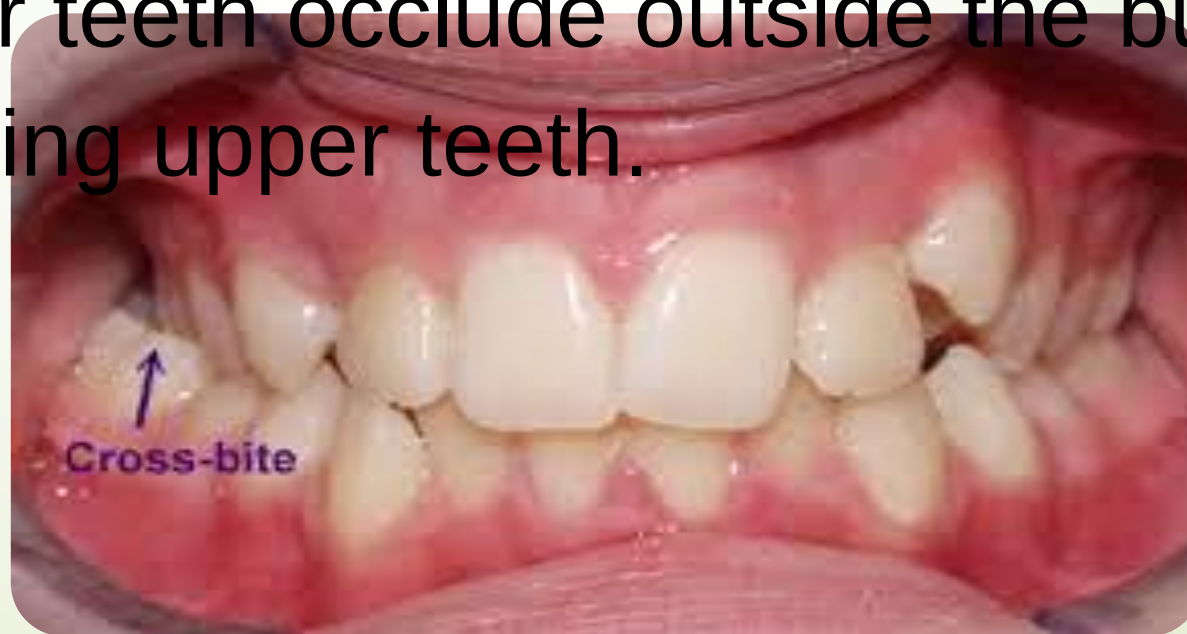
## Posterior Open bite:

- ➡ (Lateral) – open bite in between U/L posterior teeth.



## Cross bite:-

- A) Posterior cross bite - is a transverse discrepancy in arch relationship while in occlusion.
- The lower arch is wider than the upper so that the buccal cusps of the lower teeth occlude outside the buccal cusps of the corresponding upper teeth.



## B) Anterior cross bite

- ▶ antero posterior discrepancy with the U/L arches in occlusion. the lower anterior tooth or teeth occlude in front of the upper anterior tooth/teeth.
- ▶ **The anterior cross bite should be corrected immediately as early as we can in young children.**



## Scissors Bite (Telescopic Bite):

- a lingual cross bite of the lower posterior teeth.
- Note- the lower incisors are also within or lingual to the upper incisors
- Term applies to posterior teeth relation only



## Over jet:

- antero-posterior distance in the horizontal plane from the labial surface of lower incisor to the incisal edge of the most proclined upper incisor tooth.
- Normal over jet is 2→3mm. Increased over jet is found commonly with proclination of upper incisors as in



## Reverse over jet: **anterior cross bite**

- ▶ over jet found in cases where the upper incisor/s are located behind the lower
- ▶ Value is negative in mm.
- ▶ cases of anterior cross bite / Class III cases & is sometimes called 'Under bite'



## Diastema:

- Natural space present between teeth.
- A spaced deciduous dentition is also known as **‘Open dentition’**. It is an ideal and favorable feature



- A deciduous dentition without developmental spacing is called **‘Closed dentition’**. It is unfavorable and indicates future likelihood of crowding in the permanent dentition.



## ➤. Median diastema:

- Space between the upper or even the lower Central incisors in the midline of the dental arches is called median diastema or midline diastema.

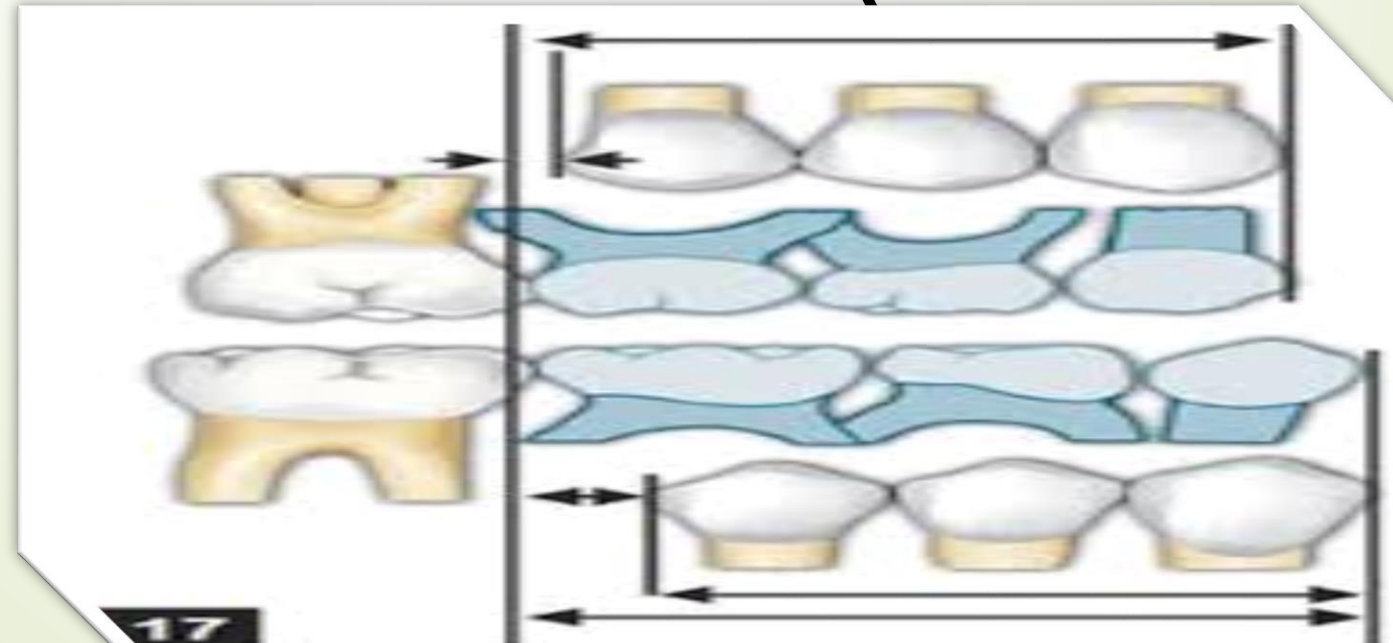


## Lee-way space:

the difference in the mesio-distal size between CDE & 345.

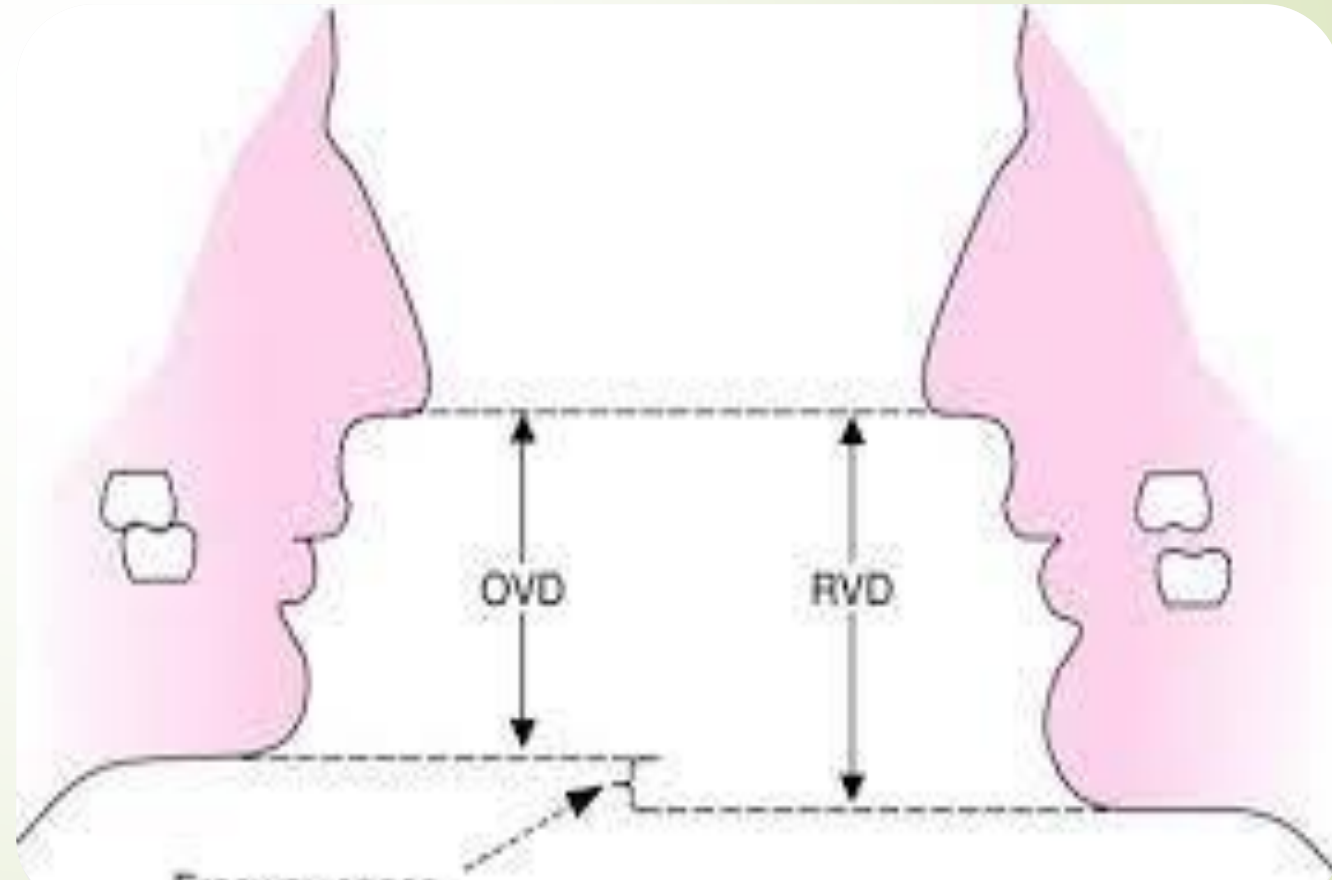
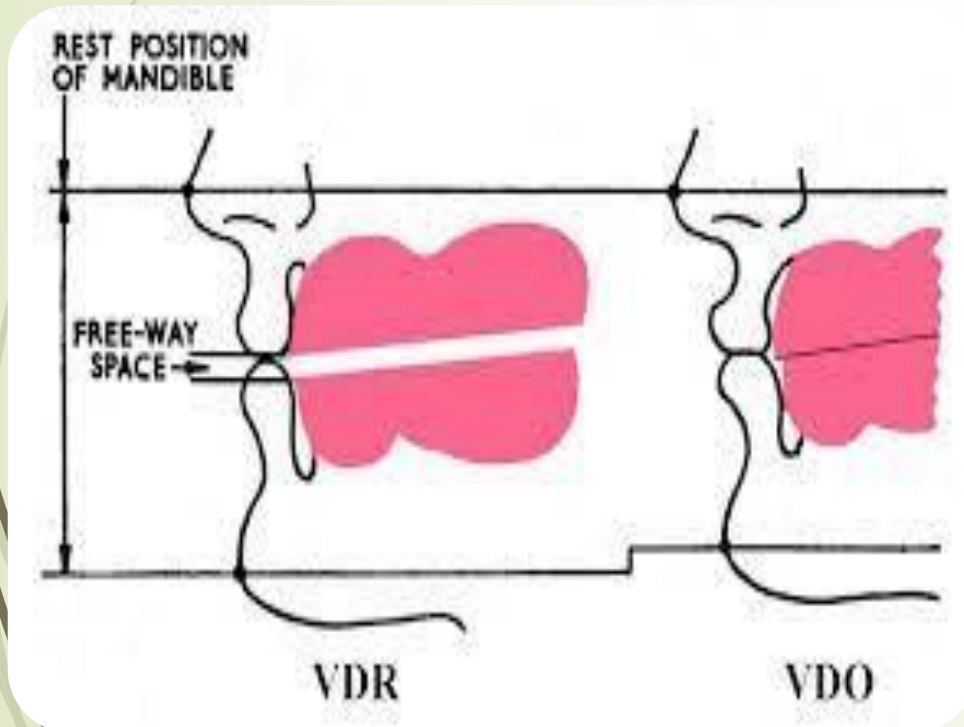
**In the upper arch it is 1.8mm (0.9mm on each side)**

**in lower dental arch it is 3.4mm (1.7mm each side).**



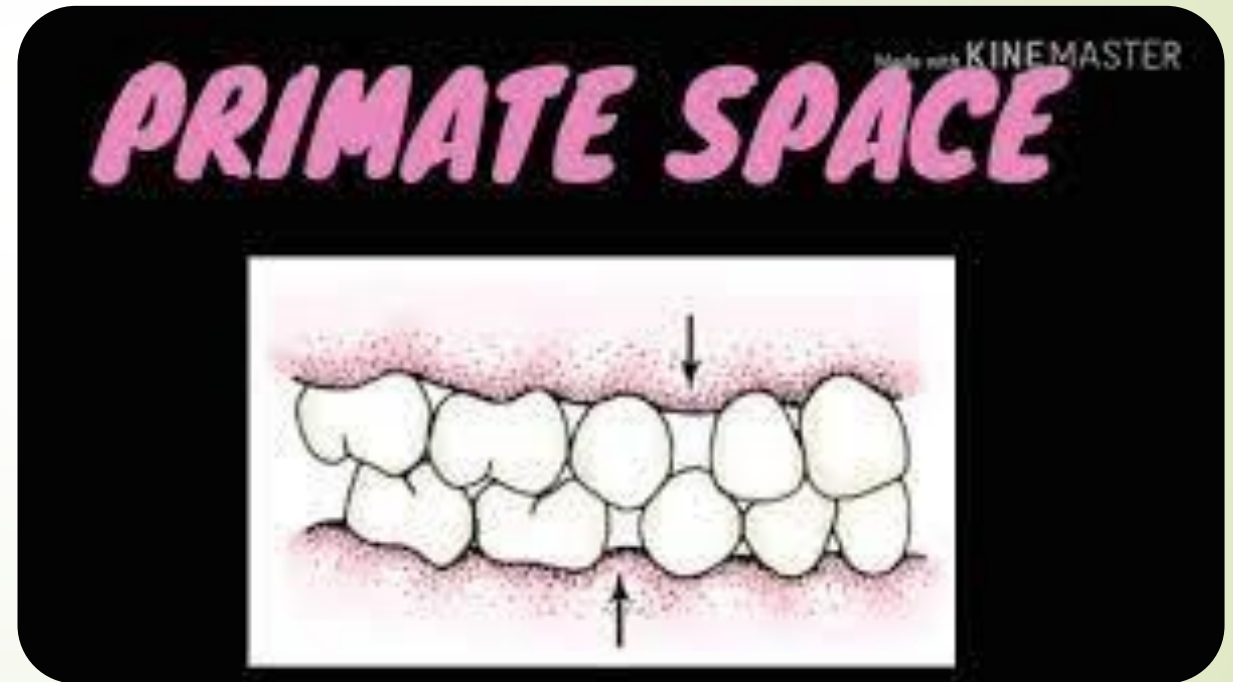
## Free-way space:

- The inter-occlusal clearance or space seen when the mandible is in the rest position. Normal freeway space is 2–3 mm.



## Primate space:

- It is a naturally occurring space in a deciduous dentition mesial to the upper canine and distal to the lower



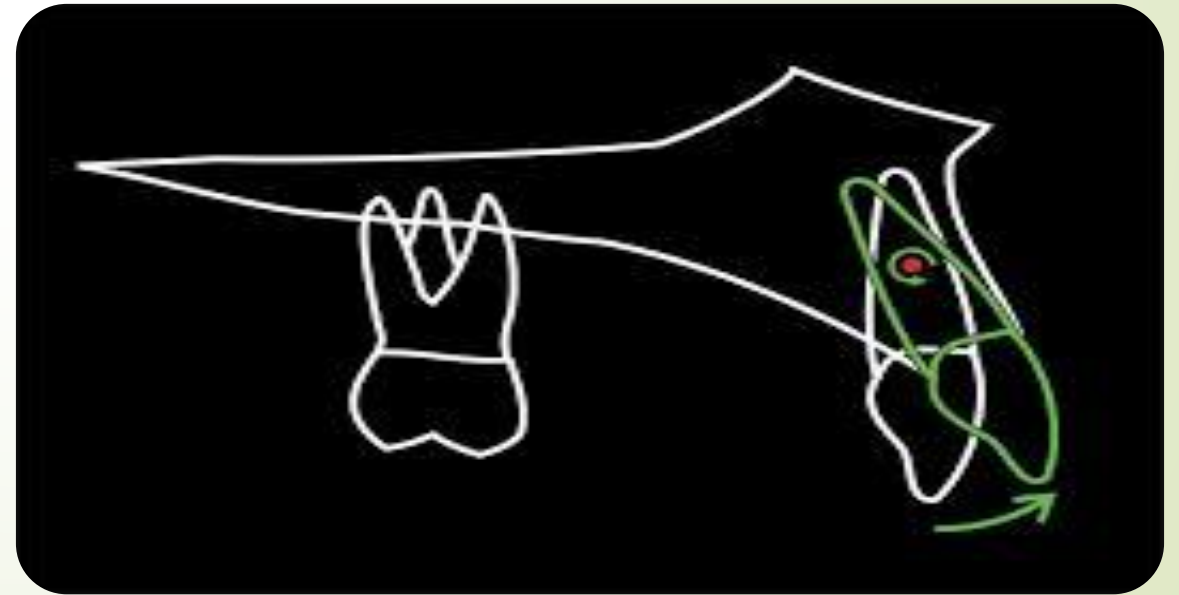
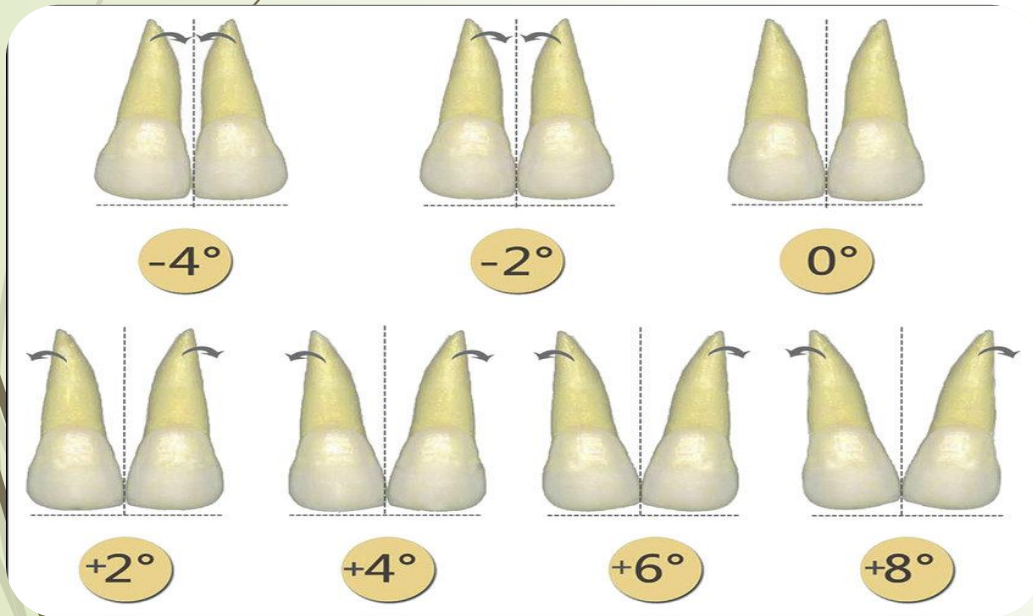
➤ **Proclination:** The labial tipping of incisor teeth.

- **Retroclination:** The lingua/palatal tipping of incisor teeth



➤ **Incisor inclination:** the degree of tip in the labio palatal plane.

➤ **Incisor angulation:** the degree of tip in the mesio-distal plane.



- **Labioversion:** The tooth tilted to the labial
- **Linguoversion:** Tooth tilted lingually.



## Fusion:

- Union of the dentine of two teeth from two tooth buds.  
radiograph will show a single crown with two roots.
- Clinically, a fusion results in one less tooth in the dental arch



## Gemination:

- a developmental anomaly due to an incomplete division of one single tooth bud.
- A cleft or notch is seen extending vertically from the incisal edge of the crown. radiograph reveals a single root & there will be no change in the number of teeth in the dental arch.



## Twinning:

- In twinning a single tooth bud divides completely into two to form two separate teeth.



# Concrescence

- : Union of the roots of two separate teeth by their cementum.



## Dilacerated Root

describes an abnormal angulation between the crown and root of a tooth or within the length of the root.



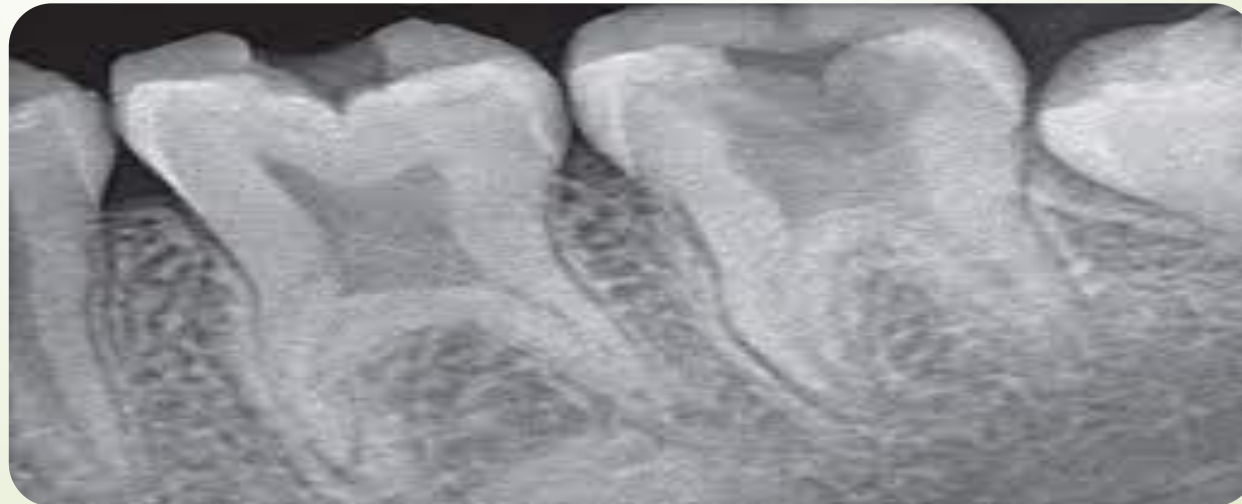
# Taurodontism

an abnormal crown-root ratio.

It almost always involves a molar tooth.

Single or multiple teeth may be affected either unilaterally or bilaterally.

Taurodontism is not diagnosed clinically but can be done only on a radiograph,



## Ectopic eruption:

- Abnormal Eruption path or position of a tooth in the dental arch.





## Transposition:

- a condition of ectopic eruption in which two teeth exchange their normal sequence or position.



## Transmigration:

- a condition where the tooth from one side is located across the midline towards the other side of the dental arch



## Retained deciduous teeth:

- Presence of a deciduous tooth beyond its normal expected exfoliation time



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- Due to this, there may be a delay in eruption of the permanent successor or result in its impaction or ectopic eruption.
  - One reason for "over-retention" is the lack of a permanent successor.



# ankylosis

- ➡ ankylosis, a condition in which the root surface becomes fused to the alveolar bone.

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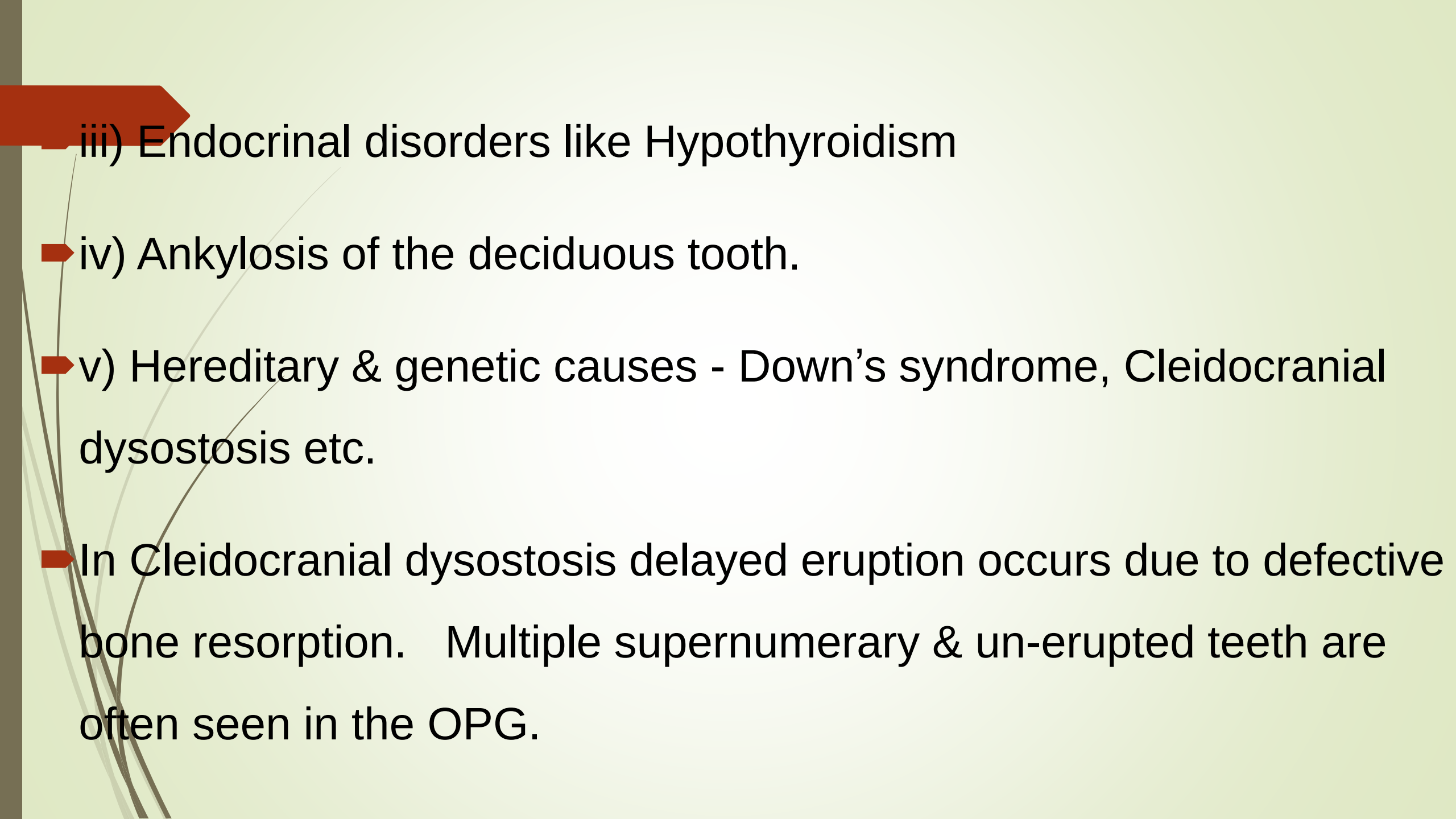
➤ The primary teeth most commonly ankylosed are the mandibular primary first and second molars,

➤ followed by the maxillary primary first and second molars.

Extraction of the ankylosed tooth will allow for the eruption of the permanent tooth.

## Delayed Eruption

- ➡ delay in eruption of a tooth beyond its normal expected time.
- ➡ Causes of delayed eruption are:-
  - ➡ i) Obstruction by supernumerary tooth and thick muco-periosteal barrier.
  - ➡ ii) Premature loss of deciduous tooth with sclerosis of bone.

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- iii) Endocrinal disorders like Hypothyroidism
  - ➡ iv) Ankylosis of the deciduous tooth.
  - ➡ v) Hereditary & genetic causes - Down's syndrome, Cleidocranial dysostosis etc.
  - ➡ In Cleidocranial dysostosis delayed eruption occurs due to defective bone resorption. Multiple supernumerary & un-erupted teeth are often seen in the OPG.

## Supernumerary teeth

- ▶ are extra teeth, more than the accepted normal number of teeth
- ▶ are more common in the upper jaw.
- ▶ Those in the midline are called 'Mesiodens'.



**➤ The supernumerary teeth behind the molars are called disto-molars &**

**➤ those adjacent to the molars are called para molars.**



➤ Multiple supernumerary teeth are also seen in :

➤ **Cleido cranial dysostosis** - a partial or complete absence of clavicles. Dental abnormalities are altered eruption patterns, over retained deciduous teeth and presence of multiple supernumerary teeth.



Fig. 2. Anterior view of the patient with cleidocranial dysplasia. Observe the prominence of the frontal and parietal bones, determining the increase of the cranial perimeter.

## Supplemental tooth:

- a supernumerary tooth of normal form, shape & size similar to the other normal tooth in morphology. It fully resembles the normal tooth.



## Missing Teeth:

**Less than normal number of teeth in the jaws.**

- Congenitally missing one or more teeth is perhaps our most common congenital malformation.
- Three terms are commonly used to describe the dental anomaly of missing teeth:-
  - a) Anodontia.      b) Oligodontia &      c) Hypodontia

## Anodontia:

- denotes total congenital absence of all teeth.
- It is a rare genetic anomaly characterized by the congenital absence of all deciduous & permanent teeth. It is associated with a group of skin and nerve syndromes called 'ectodermal dysplasias'.



## Oligodontia

is the absence of many teeth.



## Hypodontia:

- is the absence of few teeth. (Example - Upper lateral incisors)





**The Most commonly missing teeth in order of frequency are:-**

the third molars , lower second premolars and permanent upper lateral incisors .

➡ Missing laterals are slightly more common in females

## impacted tooth

- an tooth which remains un-erupted or not fully erupted due to an **obstruction** or **Lack of Space** in the dental arch. The 3<sup>rd</sup> molars and upper canines are most commonly impacted.



## Embedded tooth

- tooth which remains un-erupted within the bone due to a **Lack of Eruptive Force** & usually covered completely with bone.



## Submerged tooth:

- ***Submerged teeth*** are those that stop their relative occlusal movement in the dental arch during or after the period of active eruption.
- a retained deciduous tooth (usually a molar) which is in infra occlusion. This is because it is usually ankylosed & cannot erupt permanent tooth.



## The talon cusp

is an accessory cusp located on the lingual surface of maxillary or mandibular teeth. The cusp arises in the cingulum area and may produce occlusal disharmony. In combination with the normal incisal edge,



## Peg Lateral:

- ➔ Peg lateral is a developmental anomaly of the maxillary *lateral* incisor which resembles a small *peg* or cone shaped.
- ➔ It is present in 1→2% of the population,
- ➔ more common in the female
- ➔ can be unilateral or bilateral .
- ➔ It badly affects the aesthetics.





## Natal tooth

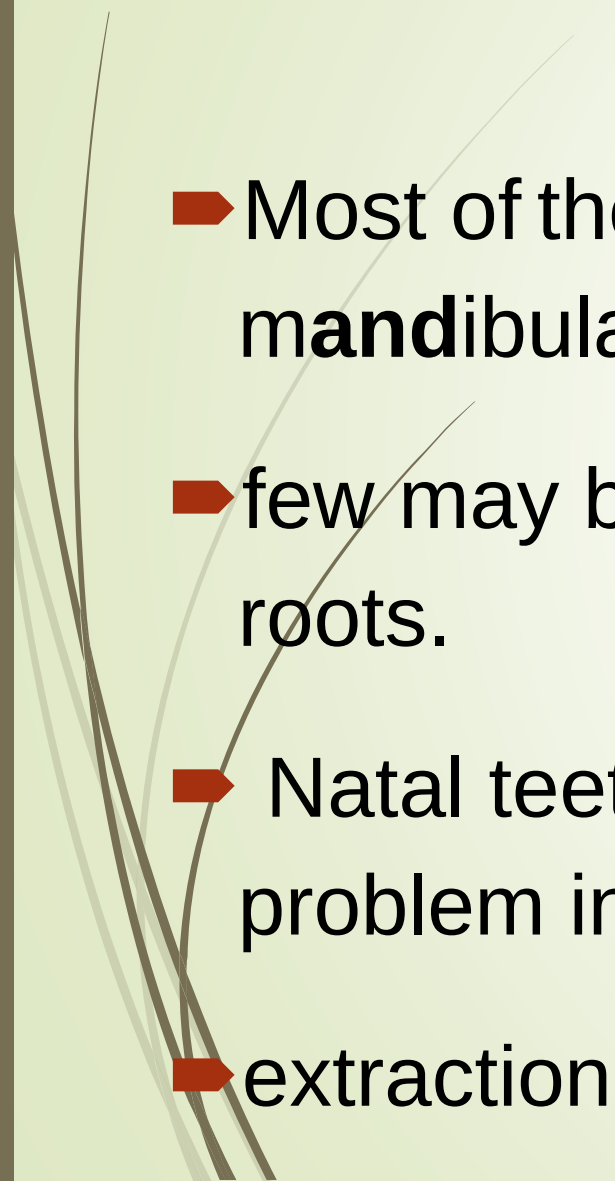
the tooth seen to be present at birth.

It is a rare anomaly with an incidence of 1 in 3,000 births.

## Neonatal tooth

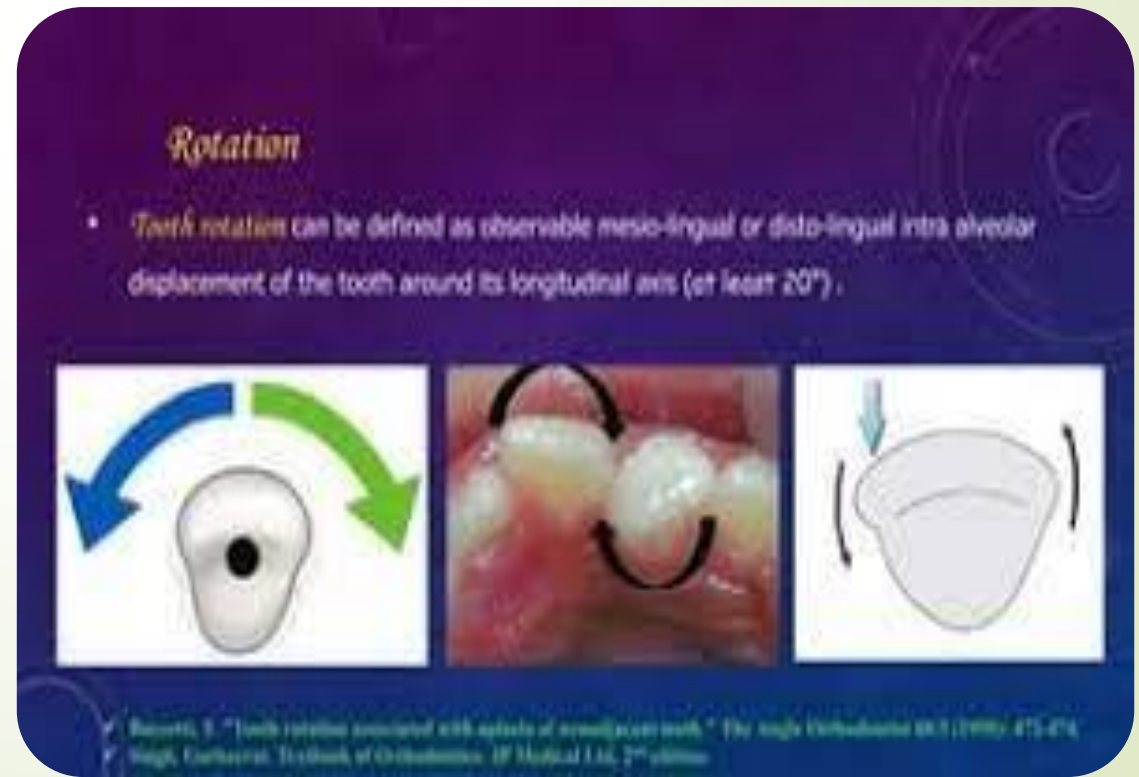
are ***teeth*** that are seen to erupt during the first month of life **Natal teeth** are more frequent than **neonatal teeth** (3:1).



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- ➡ Most of the **natal** or **neonatal teeth** are the deciduous **mandibular** incisors
  - ➡ few may become loose since they have poor or no roots.
  - ➡ Natal teeth are ignored if they do not cause any problem in suckling.
  - ➡ extraction

## Typical / Atypical rotation

- Typical' rotation is a tooth which shows Mesio-labial rotation. In 'Atypical' rotation a tooth shows mesio-lingual/palatal rotation.



## Competent lips

- - Lips which can come in contact with each other in the rest position of mandible without any strain of the muscles



## Incompetent lips

- - In the postural rest position the upper & lower lips are not able to close and come in contact with each other without some strain or hyperactivity of the lip & mentalis

